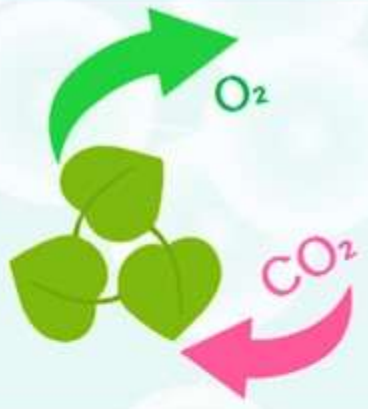




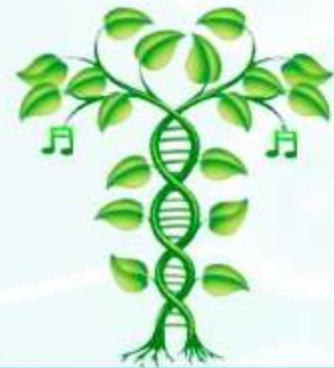
BIOLOGY



FOOD & NUTRITION-1



MODULE - 1



WHAT DO LIVING ORGANISMS EAT?



PLANTS PREPARE
THEIR OWN FOOD WITH
THE HELP OF
SUNLIGHT AIR WATER

PRODUCERS



SOME ANIMALS EAT
ONLY PLANT AND
PLANT PARTS

HERBIVORES



SOME ANIMALS EAT
THE FLESH OF
OTHER ANIMALS

CARNIVORES



HUMANS EAT BOTH
ANIMAL AND PLANT
PRODUCTS

OMNIVORES



FOOD AND NUTRITION

HOW DOES AN INDIAN MEAL LOOK LIKE?



- Food in India is very diverse across different geographical locations and across different cultures.
- An average Indian meal consists of rice, roti(bread), dal, vegetables, meat or milk products, eggs and a sweet!



FOOD INGREDIENTS

WHERE FROM WE SOURCE FOOD?

PLANT SOURCE

1 CEREALS



2 VEGETABLE



3 FRUITS



ANIMAL SOURCE

1 MEAT



2 MILK



3 HONEY



WHY DO WE NEED TO EAT FOOD ???



- 1** Our body needs constant supply of energy to perform all the functions
- 2** This energy cannot be produced by the body itself
- 3** Animals and humans need to get food from other sources
- 4** When our body gets food it breaks it down to simple forms
- 5** These simplest forms are combusted to get energy and perform all other functions in the body



WHAT MAKES THE DIFFERENCE?



A boy with higher weight
than normal



A boy with lower weight
than normal

NUTRITION





NUTRITION

DEFINITION

It is defined as the process of obtaining food necessary for health and growth



TYPES OF NUTRITION

NUTRITION

AUTOTROPHIC



HETEROTROPHIC



SAPROTROPHIC



TYPES OF NUTRITION

1

AUTOTROPHIC NUTRITION

The mode of nutrition where the organisms make the food themselves from simple inorganic substances

The organisms are called autotrophs



2

HETEROTROPHIC NUTRITION

The mode of nutrition in which organisms take ready made food prepared by other organisms

The organisms are called heterotrophs



3

SAPROTROPHIC NUTRITION

The mode of nutrition in which the organisms take in nutrients in solution form from dead and decaying matter
The organisms are called saprotrophs



TYPES OF NUTRITION

1

AUTOTROPHIC NUTRITION



2

HETEROTROPHIC NUTRITION



3

SAPROTROPHIC NUTRITION



TYPES OF NUTRITION

1

AUTOTROPHS

PHOTOAUTOTROPHS

- 1 Receives energy from sunlight and convert into food materials
- 2 Photoautotrophs are usually found on the surface of Earth where sunlight can reach



CHEMOAUTOTROPHS

- 1 Receives energy from chemicals and inorganic substance
- 2 Chemoautotrophs are usually found under the surface of Earth where sunlight cannot reach



TYPES OF NUTRITION

1

AUTOTROPHS

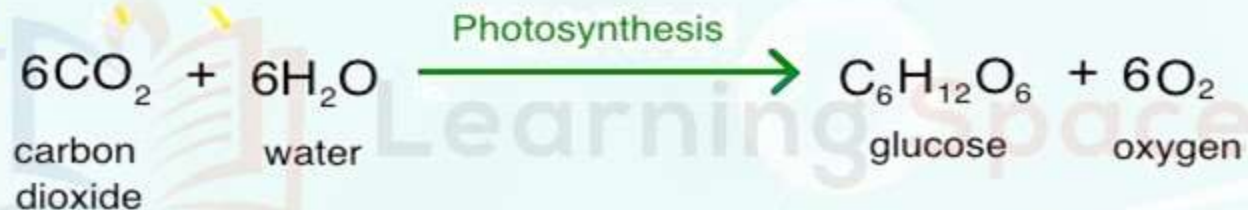
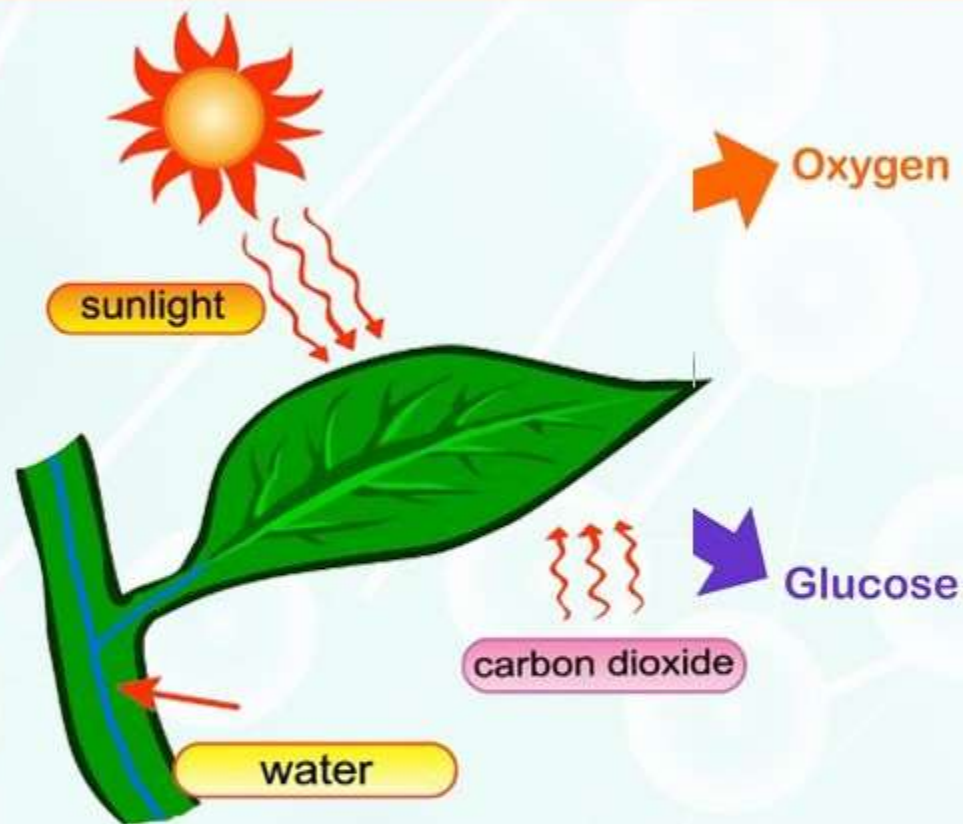
PHOTOAUTOTROPHS



CHEMOAUTOTROPHS



PHOTOAUTOTROPHS MAKE FOOD BY A PROCESS CALLED PHOTOSYNTHESIS



NUTRITION TYPE - AUTOTROPHIC

PHOTOAUTOTROPHS

GREEN PLANTS



ALGAE



CYANOBACTERIA



NUTRITION TYPE - AUTOTROPHIC

CHEMOAUTOTROPHS

ARCHEA BACTERIA



SULPHUR AND IRON BACTERIA



NITROGEN BACTERIA



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TYPES OF NUTRITION

KINDS OF HETEROTROPHIC NUTRITION

HOLOZOIC NUTRITION

The mode of nutrition where the organisms internalize or ingests the food particle and process them. All animals and amoeba are holozoic



PARASITIC NUTRITION

The mode of nutrition in which the organism takes nutrition from living source and deprives the host of its valuable nutrition. E.g cuscuta (amarbel)



NUTRITION TYPE - HETEROTROPHIC

HOLOZOIC NUTRITION

1

HERBIVORE



Feed on algae and other plant materials

2

CARNIVORE



Feed on flesh of other animals

3

INSECTIVORE



Feed on insects



NUTRITION TYPE - HETEROTROPHIC

HOLOZOIC NUTRITION

4

FRUGIVORE



Feed on only fruits

5

SANGUINIVORE



Feed on the blood of other animals

6

DETRIVORE



Feed on dead and decaying matter



NUTRITION TYPE - HETEROTROPHIC

HOLOZOIC NUTRITION

CARNIVOROUS/ INSECTIVOROUS PLANT



Pitcher plant have modified leaves to trap insects which sticks in the inside



Venus flytrap forms a snap trap which rapidly closes when insects sit on it



Sundews Digests insects using mucilage secreting glands



NUTRITION TYPE - HETEROTROPHIC

PARASITIC NUTRITION

TYPES OF PARASITES

ECTOPARASITES

Parasites who feed on the external surface of the host organism.

- E.g. Lice on humans
- Ticks and fleas on dogs
- *Cuscuta* (Amarbel) on plants and trees.



ENDOPARASITES

Parasites who live inside the host's body at different organ sites.

- E.g. Roundworms in the intestines.
- *Plasmodium* in RBC
- *Leishmania* in liver and bone marrow. (causes Kala azar)

TYPES OF PARASITES

ECTOPARASITES



Louse



Tick

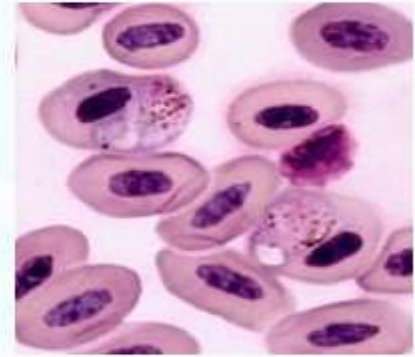


Flea

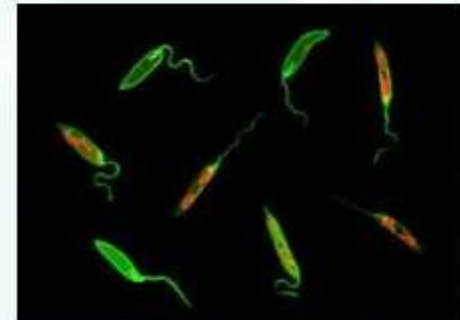
ENDOPARASITES



Roundworm



Plasmodium



Leishmania



NUTRITION TYPE - HETEROTROPHIC

PARASITIC NUTRITION

COMMON PLANT PARASITE



RAFFLESIA sp.
The world's largest flower,
smells of rotten meat found
In Southeast Asia



CUSCUTA, a.k.a amarbel is
a yellowish parasite found
in tropical regions have
medicinal properties



MISTLETOE grows in
colder climate of Europe
and also called as the
symbol of Christmas



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NUTRITION TYPE - SAPROTROPHIC

FUNGUS AS SAPROTROPHS



Mucor fungus or commonly found in soil or harvested crops



Rhizopus fungus commonly called as bread mould

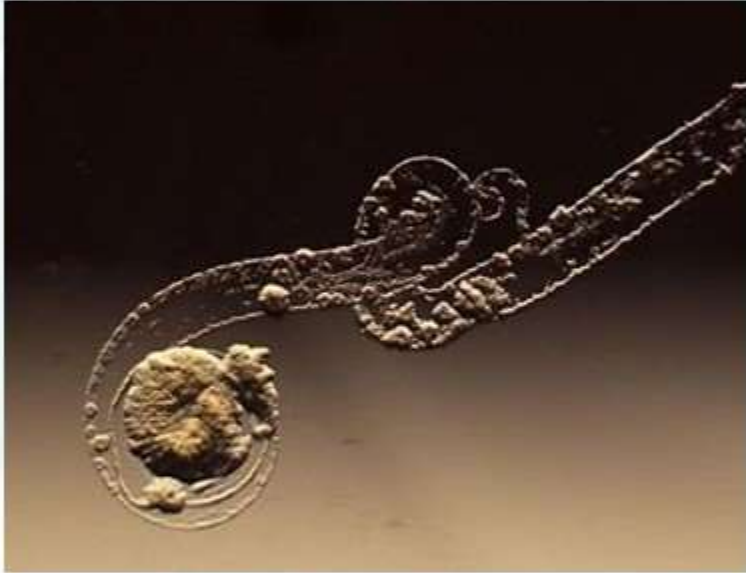


Mushroom grows on food surface and wood and is a great medicinal and food source

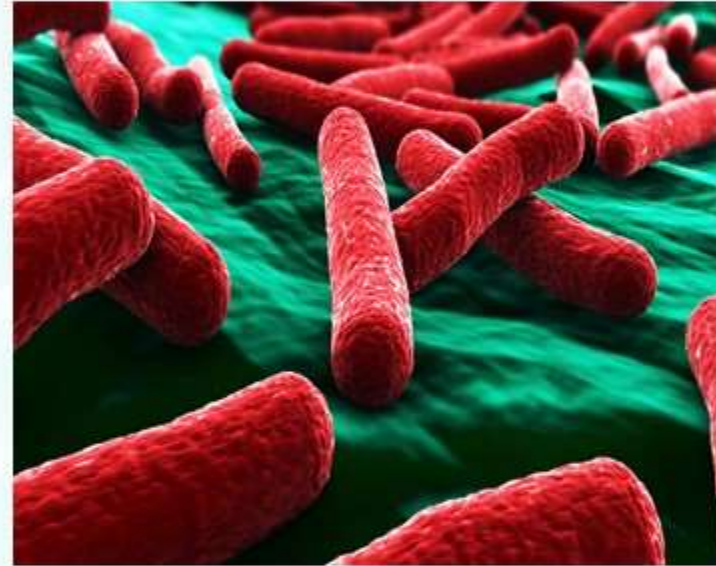


NUTRITION TYPE - SAPROTROPHIC

BACTERIA AS SAPROTROPHS



Spirochaeta cytophaga
has the capacity to
decompose cellulose



Escherichia coli has the
capacity to decompose
animal flesh



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OUR TEAM

G. V. Rao	Amrita Naidu
D. Sunil Kumar	G. Ravi Babu
V. Dayakar James	D. Joji
Y. Sri Poojita	K. Victor Babu

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